Q20 Discussion Questions

1. The skeleton (SGD + backprop + tensor hardware) has been stable for a decade. What would it take for a genuinely non-skeleton architecture to compete — and do state-space models or diffusion LMs qualify, or are they skeleton-loyal?
2. "Transfer of understanding": which of your CNN-era intuitions have you personally seen fail on LLMs — and which saved you?
3. MoE sparsity reuses a 1990s gating idea at 2020s scale. What other "old" ideas do you expect to be recycled next, and why does recycling keep working?
4. The post-training stack (SFT, DPO/RLHF, RLVR) has no analogue in vision models. Is that a fundamental difference in kind, or an artifact of language being easier to verify than pixels?
5. If LLM internals are mostly inherited parts, why did the "scaling race" favor Transformers specifically? What property of the architecture made it the winner, and could that property have been identified in 2017?